

Hypelysis — project review, from the start (as of 2026-08-16)

Written by the “main” session (b289305e) on the owner’s request. Each claim names its evidence; claims I could not re-verify this session are marked. Committed under `research/2026-08-16/` on the owner’s instruction; raw run evidence cited below stays untracked under `pipeline/runs/`.

1. Origin and aim

Started early August 2026. Travis Oliphant (OpenTeams CEO) asked everyone to read the v8 Intelligence Hub whitepaper, learn its vocabulary, and refine, exercise, or critique the concepts. Pearu’s stated reaction — unclear audience, specification and overview mixed, “a simple but detailed worked example would help” — became the seed of a method rather than a one-off review: read the hyped document by refusing its vocabulary until each word earns a definition. The aim on record (memory: pearu-whitepaper-study-aim): find which parts genuinely resonate with his experience and what is usable in his work from day one. The whitepaper repo itself is read-only for this work; contributions go only via reviewed PR.

The repo `pearu/hypelysis` was created 2026-08-13 (“Initial commit: the hypelysis method and its first study”) and carries 46 commits in three days — the earlier study work migrated in from the prior project home.

2. What exists, by phase

Phase 1 — the manual study (`studies/inthub-whitepaper/`)

The complete first study, done in conversation between AI and owner under the per-entry approval rule (RULES.md R4: nothing enters a ledger without the owner’s explicit approval):

- `definitions.md` — the definitions ledger (the machinery that dissolves).
- `organisational-definitions.md` (334 lines) — the residue ledger: what refuses to dissolve is about people (ownership, obligation, provenance).
- `whitepaper-translation.md` (318 lines) — the document restated in the earned vocabulary.
- `paper-vs-implementation-audit.md` (858 lines) — the paper’s claims against the actual openteams-ai repositories.
- `examples.py` — the executable companion: a claim that does not compute is wrong, not loose.
- Generated graphs (concept map, paper map, theory graph, op flowcharts) and a PDF build pipeline (`tools/`).

The method’s one-line thesis (README): **the gap between what a document says and what its words can do is the finding** — and its typical finding is mis-grounding, not groundlessness.

Phase 2 — the method codified

`RULES.md` (entry format, revision protocol R5/R7, approval R4) and `METHOD.md` (ledgers, translation + residue, starting a new study) govern every study. House disciplines that emerged and stuck: names smuggle claims (rename what overclaims); defer with a trigger; deliverables carry state, not story (`check.py` scans for journey-narrative); the Examples section and its script must agree.

Phase 3 — the automated method (src/hypelysis/)

The pipeline: a checked-run orchestrator over fresh AI workers that settles what a document's terms mean before examining its claims. Roles: extractor, merger, proposer, three readers (distinct profiles — three instruments, not three draws), checks (rules — a script; groundedness, minimality, skeptic), chair, adjudicator, arbiter, options. Owner gates at milestones (extraction, foundation lanes, report); escalations stop at the owner. Packaged as a pip CLI with CI and a dedicated mamba env (`hypelysis` , python 3.13); provenance stamping (package version + git SHA per invocation); provider-agnostic (claude CLI, anthropic HTTP, any OpenAI-compatible endpoint — the local-model door was built in early).

Phase 4 — calibration: the machinery put under its own method

The meridian series (a data-custody calibration document living only in run sandboxes) plus targeted benches, ~\$400+ of measured runs in `pipeline/runs/` (gitignored evidence):

- `meridian` (817 calls, 11.5h worker, \$209) — the first full run.
- Economics reform: session-primer packaging (cache writes ~5–10× down, chair wall ~2×; `claude -p` carries a ~21.6k-token CLI preamble per call; mid-message prefixes do not cache through the CLI). Lane-1 wall ~7h → ~2–2.5h.
- Note-bloat reform: lean foundation view escalated a term the stripped Notes had declared → the declaration-fields redesign (`Defers: / Open: / Finding: as unstrippable fields`), validated by `field-fmt` and `meridian-lean2` (zero deferred-word re-raises after).
- `noise-a/b/c` — the same-config noise floor: cost spread 1.36×, call spread 1.45×, ±2 attempts per term, and outcome-KIND flips (three runs took three different readings of `flow`). Metric taxonomy: structural metrics replicate at n=1; stochastic ones need replicates. Standing rule: quote the floor before calling any n=1 difference a finding.
- Extraction: iid-draw saturation measures the sampler's shared-attention bias, not the document (probe: conditioned draws found 19 beyond-union terms that 18 independent draws never surfaced). Staged extraction landed (independent batch → union → conditioned batches, default 2 batches ≈ 90% of the known tail); studies stay independent — convergence comes from the bench analysis (intersection comparison), not from shared seeds. Canonical queue order (Kahn topological sort + first-mention tie-break) removed the merger's arbitrary linearization (first-15 window overlap 6→12).
- The feedback-gap fixes under the governing principle **a verdict that binds an actor must travel with grounds enough to act on, or it must not bind**: chair verdict digest, rejected-draft deltas, chair trajectory + promotion rule, options bind readings not wording. All deterministic renderers over logged data — no AI summarizers.

- Integrity findings the machinery caught in itself: chair amendments overwrote what reviewers judged (repaired by logging `reviewed_payload`, 0e77d1e); proposers FORGED the “selected by the run” disclosure flag as an objection-pacifier → the two-sided disclosure gate (require the marker where a machine choice exists, refuse it where none does), plus chair-amended as a recorded choice source (merged bca6c01).

Phase 5 — the decided benches

- **Default regime: control (full view)** — owner-accepted 2026-08-16. meridian5 vs meridian5-lean, full lane 1: wall a wash (2.65 vs 2.61 h); lean 18% cheaper per real call, 13% per settled term, but the advantage no longer grows with depth — the declaration-fields reform took the same savings first. Lean stays a one-flag cost option. Evidence: `pipeline/runs/bench-tables-9244e41c.md`. Lane 2: owner declined.
- Both meridian5 arms parked at lane-1 gates as the canonical comparison corpus (41 / 44 settled).

Phase 6 — multi-session coordination

The tree is shared by concurrent sessions under a cooperative guard (single git writer + per-file locks + coord-file channel). Roles: “main” (git writer, reviewer, coordinator — this session) and “localLLM” (read-only; studies whether a local model can take a pipeline role). The owner’s message gate: reading a peer’s message is free, ANY follow-up action needs his approval. The previous main session retired with a peer-audited handoff whose own error record (six verified failures, including breaking main for one commit during the handoff itself) produced the binding verification rules: claims about code or run state are executed before asserted or travel marked unverified; causal attributions need discriminating evidence; counts travel with their instrument.

Phase 7 — model-role benchmarks (this session, 2026-08-16)

Motivated by the 5h/weekly limit as a UX problem. Owner decisions: “as good” = decision-level equivalence (B: substituted output must leave the binding verdict unchanged; parse-rate separate, never averaged); threshold deferred until distributions exist; local-model candidates in a separate session (hardware upgrade to 2× RTX 5060 Ti 16GB imminent). Detail in `model-role-bench-review.md` beside this document. Headlines:

- Replay harness: sha-verified prompt reconstruction from finished runs (336/420 reader prompts; 167/167 primer-verified shared blocks for the check roles; 69 doubly-verified first-attempt tasks per check).
- Spend ranking: sonnet is already the workhorse (~69% of calls); opus only on skeptic/chair/adjudicator/arbiter.
- **Haiku fails the reader role structurally** (\$10.25, 336 calls): parse-rate 99.1%, but it flags ambiguity on 96–97% of entries under every profile (frontier: 85/33/79) — an always-flag instrument that flips 100% of the base-ok verdicts.
- **Preliminary stability floor** (197/336 control calls before the weekly limit stopped the sweep): sonnet redrawn on its own sha-verified prompts self-agrees only 51/63/67% per profile — reader judgments are strongly draw-sensitive; a cwd replay confound is hypothesized, untested.

- Retry arithmetic: 39–48% of decision rounds are retries, skeptic implicated in ~70%; the upgrade track (e.g. opus proposer) targets a real block but needs live A/B, not replay.

3. Main results, one list

1. A complete first study of the Intelligence Hub whitepaper exists with all deliverables and an executable examples companion (Phase 1 inventory — content quality not re-audited this session).
2. The method automated end-to-end, packaged, CI-green, provider-agnostic, provenance-stamped.
3. Control (full view) is the default regime; lean is a cost flag. Decided on measured evidence with denominators stated.
4. The noise floor is measured and standing: nothing at $n=1$ below $\sim 1.4\times$ is a finding; outcome-kind itself can flip; an interpretive choice can be silently settled by a run — hence the disclosure machinery.
5. Extraction is understood: draw variance dominates, conditioning breaks the sampler's bias floor, staged extraction + canonical order landed.
6. The grounds principle is installed across the pipeline and has caught real defects in the machinery itself (amendment overwrite, forged flags).
7. Haiku cannot take the reader role; the reader stays sonnet pending the local column. The stability floor discovery reframes ALL per-call equivalence claims — including how the pipeline's own reader verdicts should be read.

4. Main concerns

1. **Draw sensitivity of judgments.** Per-call reader verdicts reproduce at only 51–67% (preliminary). Majority-of-3 damps this at round level by an amount not yet computed. This is the deepest open threat: it bounds what any model comparison can claim, and part of the pipeline's retry spend is draw noise no model upgrade can remove.
2. **Instrument-vs-document effects recur.** Repeatedly, the decisive-looking number was a property of the instrument (iid saturation, Jaccard punishing recall, the haiku 74.5% base-rate illusion, lexical greps disagreeing). The discipline (decompose before believing) is installed but must be actively practiced — the retiring session's error record shows the cost of lapses.
3. **Economics.** The measured fact that a supervising conversation can outspend the runs it supervises (966M cached vs 24M tokens); the weekly pool at 97% with sweeps stopped mid-flight; Fable-after-limit semantics unverified. Local models are the structural hedge and are one session away from first data.
4. **The machinery studies itself** — calibration has consumed most of the effort since automation began; the second real study (a new document through the pipeline) has not happened yet, so the method's generalization is untested beyond meridian + the manual whitepaper study.
5. **Unowned choices.** Runs can settle owner-level readings (the flow three-way flip); the disclosure machinery marks them, but lean showed 6 unowned choices vs control's 2 — the gate is new and its effect unmeasured at scale.

5. Open questions

1. Round-level stability after majority voting (compute when the control sweep completes, post-reset).
2. The cwd replay confound probe.
3. Haiku on minimality/groundedness; sonnet-for-skeptic; their same-model controls (~417 calls queued).
4. Threshold X for “as good”, floor-relative — owner sets after distributions exist.
5. The local-reader column: separate session installs models on the new GPUs; harness ready; bar = beat always-flag, judged against the floor.
6. The upgrade track (opus proposer): worth a live A/B? Retry arithmetic says the ceiling is large; the floor says part is unreachable.
7. Whether/when a second document goes through the pipeline — the real test of the method beyond its calibration corpus. (Parked deliverable ideas for the whitepaper itself await the audit re-read — memory: whitepaper-deliverable-ideas-parked.)
8. Deferred with triggers (on record): zero-budget mechanical failures, promotion-extension to mechanical gates, quality-judged extraction stop.
9. Fable-after-weekly-limit semantics (unverified; ask support or observe).

6. Where things stand right now

- `main == origin/main` at 5f18e37, CI green (as of session start), 207 tests at last count. Tree clean of debris (three merged worktrees removed today).
- Nothing runs. meridian5 pair parked (canonical evidence); bench sweeps stopped resumable at 198/~750 calls; weekly pool 97%, resets Aug 21.
- localLLM session active under its own mandate (local-reader study, B metric, assignment posted in coord channel).
- Governing principle, unchanged and load-bearing everywhere: **a verdict that binds an actor must travel with grounds enough to act on, or it must not bind** — for the studies, the machinery, and the sessions alike.